Non-Linear Optimization

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Part I
Vector Calculus

1 Functions

A **function**, $f : \operatorname{dom} f \to \operatorname{cdm} f$, is a mapping which associates each element in a set, $\operatorname{dom} f$, to one, and only one element in another set, $\operatorname{cdm} f$.

We call $\operatorname{dom} f$ and $\operatorname{cdm} f$, the **domain** and **co-domain** of f, respectively, both of which, in general, are vector spaces of arbitrary dimensions and over arbitrary fields.

For our purposes though, we will assume these fields are the same, namely, the set of real numbers, $\mathbb{R}$. Further, we will focus on only **scalar-valued functions**[1], i.e., functions whose co-domain is 1-dimensional.

While we will still assume the dimension of the domain is arbitrary, we will independently consider two cases:

1. when the domain is 1-dimensional, i.e., that of a **univariate** function, and
2. when the domain is n-dimensional, $n > 1$, i.e., that of a **multivariate** function.

We begin with a few basic definitions and theorems from calculus.

[1] A function whose co-domains is n-dimensional, with $n > 1$ is said to be **vector-valued**.

1.1 Univariate Scalar-Valued Functions on $\mathbb{R}$

We will begin with real scalar-valued univariate functions over the real numbers. For simplicity, we will refer to any such function as simply being a "real function".

With that being said, let's consider a real function, $f : \mathbb{R} \to \mathbb{R}$.

1.1.1 Continuity, Differentiability, and Smoothness

We say f is **continuous** at a point, $x_0 \in \mathbb{R}$, iff

$$\lim_{x \to x_0} f(x) = f(x_0).$$

If f is continuous at every point in $\mathbb{R}$, we say f is a **continuous function**.

We say f is **differentiable** at a point, $x_0 \in \mathbb{R}$, iff

$$\lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

exists.

When it exists, the above limit is called the **derivative** of f, at x_0, denoted $f^{(1)}(x_0)$, and gives the slope of the line tangent to f at x_0.

If f is differentiable at every point in $\mathbb{R}$, we say f is a **differentiable function**[2]. The derivative of such a function is itself a function.

In general, the nth-derivative of f, denoted $f^{(n)}$, is the derivative of $f^{(n-1)}$, assuming $f^{(n-1)}$ is differentiable[3].

We say f is a **class-n function**, i.e., $f \in C^n$, if f is at least n-times differentiable, that is, if $f^{(i)}$ exists for all $1 \leq i \leq n$, and $f^{(n)}$ is continuous.

If $f \in C^\infty$, we say f is a **smooth function**.

[2]Differentiability implies continuity.

[3]Often, f' and f'' are used to denote the first and second derivatives of f, respectively.

1.1.2 Taylor's Theorem

In general, **Taylor's theorem**[4] states that for any $f \in C^\infty$,

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k,$$

for any x_0.

We may always replace x with $x + \Delta x$, and also choose $x_0 = x$, giving an alternative form of Taylor's theorem,

$$f(x + \Delta x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x)}{k!}(\Delta x)^k.$$

The proof of Taylor's theorem is complicated, and so we consider two special cases of this latter form:

For any $f \in C^1$, where $f : \mathbb{R} \to \mathbb{R}$,

$$f(x + \Delta x) = f(x) + f'(x)\Delta x + O(\Delta x).$$

Proof:
Observe that the above statement is equivalent to the statement,

$$f(x + \Delta x) - f(x) - f'(x)\Delta x = O(\Delta x).$$

We have

$$\lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x) - f'(x)\Delta x}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{f'(x + \theta\Delta x)\Delta x - f'(x)\Delta x}{\Delta x}$$
$$= \lim_{\Delta x \to 0} [f'(x + \theta\Delta x) - f'(x)]$$
$$= \lim_{\Delta x \to 0} f'(x + \theta\Delta x) - f'(x)$$
$$= f'(x) - f'(x) \because f \in C^1$$
$$= 0$$

■

This is the 1st-order case of Taylor's theorem, i.e., when $k = 1$.

[4] When working with polynomials, we typically let $0^0 := 1$.

For any $f \in C^2$, where $f : \mathbb{R} \to \mathbb{R}$,

$$f(x + \Delta x) = f(x) + f'(x)\Delta x + \frac{1}{2}f''(x)\Delta x^2 + O(\Delta x^2).$$

Proof:
Observe that the above statement is equivalent to the statement,

$$f(x + \Delta x) - f(x) - f'(x)\Delta x - \frac{1}{2}f''(x)\Delta x^2 = O(\Delta x^2).$$

We have,

$$\lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x) - f'(x)\Delta x - \frac{1}{2}f''(x)\Delta x^2}{\Delta x^2}$$

$$= \lim_{\Delta x \to 0} \frac{\frac{1}{2}f''(x + \theta\Delta x)\Delta x^2 - \frac{1}{2}f''(x)\Delta x^2}{\Delta x^2}$$

$$= \lim_{\Delta x \to 0} \left[\frac{1}{2}f''(x + \theta\Delta x) - \frac{1}{2}f''(x)\right]$$

$$= \frac{1}{2}\left(\lim_{\Delta x \to 0}[f''(x + \theta\Delta x)] - f''(x)\right)$$

$$= \frac{1}{2}(f''(x) - f''(x)) \because f \in C^2$$

$$= 0$$

■

This is the 2nd-order case of Taylor's theorem, i.e., when $k = 2$.

1.2 Multivariate Scalar-Valued Functions on $\mathbb{R}$

We will now consider real scalar-valued multivariate functions over the real numbers. Again, for simplicity, we will refer to any such function as simply being a "real multivariate function".

With that being said, let's consider a real multivariate function, $f : \mathbb{R}^n \to \mathbb{R}$.

1.2.1 Continuity, Differentiability, and Smoothness

We say f is **continuous** at a point, $\mathbf{x}_0 \in \mathbb{R}^n$, iff

$$\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0).$$

If f is continuous at every point in $\mathbb{R}$, we say f is a **continuous function**.

We say f is **directionally differentiable** at a point, $\mathbf{x}_0 \in \mathbb{R}^n$, in the direction, $\mathbf{u} \in \mathbb{R}^n$, iff

$$\lim_{k \to 0} \frac{f(\mathbf{x}_0 + k\mathbf{u}) - f(\mathbf{x}_0)}{k}$$

exists.

When it exists, the above limit is called the **directional derivative** of f in the direction, $\mathbf{u}$, at $\mathbf{x}_0$, denoted $f^{(1)}_{\mathbf{u}}(\mathbf{x}_0)$, and gives the slope of the hyperplane tangent to to f at $\mathbf{x}_0$ in the direction $\mathbf{u}$.

Suppose there exists a vector-valued function, ∇f, so that

$$\nabla f(\mathbf{x}) \cdot \mathbf{u} = f^{(1)}_{\mathbf{u}}(\mathbf{x}).$$

We call this function the **gradient** of f.

Observe that if $\mathbf{u}$ is chosen to be the ith standard basis vector, $\mathbf{e}_i$, then

$$\nabla f(\mathbf{x}) \cdot \mathbf{e}_i = f^{(1)}_{\mathbf{e}_i}(\mathbf{x}) = \lim_{k \to 0} \frac{f(\mathbf{x} + k\mathbf{e}_i) - f(\mathbf{x})}{k}$$

$$= \lim_{k \to 0} \frac{f(x_1, \ldots, x_{i-1}, x_i + k, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_n)}{k}$$

$$:= \frac{\partial f}{\partial x_i}(x_1, \ldots, x_n)$$

$$= \frac{\partial f}{\partial x_i}(\mathbf{x}).$$

We call $\partial f/\partial x_i$ the **partial derivative** of f w.r.t., x_i.

Therefore, in rectangular coordinates, defined by the standard basis, $\mathbf{e}_1, \ldots, \mathbf{e}_n$, the gradient may be expressed as[5]

$$\nabla f(\mathbf{x}) = \frac{\partial f}{\partial \mathbf{x}},$$

where

$$\frac{\partial f}{\partial \mathbf{x}} := \begin{bmatrix} \frac{\partial f}{\partial x_1} & \cdots & \frac{\partial f}{\partial x_n} \end{bmatrix}.$$

We say f is **differentiable** at a point $\mathbf{x}$, if $\nabla f(\mathbf{x})$ is defined, that is, if every partial derivative of f exists at $\mathbf{x}$.

If f is differentiable at every point in $\mathbb{R}^n$, we say f is a **differentiable function**[6].

The derivative of a multivariate function is a vector-valued function.

We may differentiate the gradient element wise to obtain

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial f}{\partial x_1 x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_n \partial x_1} & \cdots & \frac{\partial f}{\partial x_n x_n} \end{bmatrix}.$$

We call $\nabla^2 f(\mathbf{x})$ as the **Hessian**[7] of f.

In general, the nth-derivative of f, denoted $\nabla^n f$, is an n-th order tensor, namely, the derivative of $\nabla^{n-1} f$, assuming $\nabla^{n-1} f$ is differentiable.

We say f is a **class-n function**, i.e., $f \in C^n$, if f is at least n-times differentiable, that is, if $\nabla^i f$ exists for all $1 \leq i \leq n$, and $\nabla^n f$ is continuous.

If $f \in C^\infty$, we say f is a **smooth function**.

[5] We are using the convention where $\partial f/\partial \mathbf{x}$ is a row-vector and $\partial \mathbf{f}/\partial x$ is a column-vector.

[6] Again, differentiability implies continuity.

[7] The Hessian of a function is the Jacobian of its gradient.

Note that $\nabla f(\mathbf{x})$ is the rate of change of f at $\mathbf{x}$, in the direction of steepest ascent.

We can see this by considering the definition of the dot-product.

For any $\mathbf{u}$ where $|\mathbf{u}| = 1$,

$$\nabla f(\mathbf{x}) \cdot \mathbf{u} = f_{\mathbf{u}}^{(1)}(\mathbf{x}).$$

Thus, $|f_{\mathbf{u}}^{(1)}(\mathbf{x})| = |\nabla f(\mathbf{x}) \cdot \mathbf{u}| = |\nabla f(\mathbf{x})||\mathbf{u}| \cos\theta = |\nabla f(\mathbf{x})| \cos\theta$, which is maximized when $\theta = 0$.

1.2.2 Clairnaut's Theorem

Clairnaut's theorem states that if $f \in C^2$, then $\nabla^2 f$ is symmetric. We omit the proof.

1.2.3 The Mean-Value Theorem

Suppose $f \in C^1$ is a real multivariate function. Then,

$$f(\mathbf{x} + \mathbf{v}) = f(\mathbf{x}) + \nabla f(\mathbf{x} + \theta \mathbf{v})\mathbf{v}.$$

This is the 1st-order case.

Proof:
Let $g(t) := f(\mathbf{x} + t\mathbf{v})$.

Observe that $g \in C^1$ is a real univariate function.

From the mean-value theorem for univariate functions,

$$g(0 + 1) = g(0) + 1g'(0 + \theta \cdot 1)$$

$\Rightarrow g(1) = g(0) + g'(\theta)$
$\Rightarrow f(\mathbf{x} + \mathbf{v}) = f(\mathbf{x}) + g'(\theta)$
$= f(\mathbf{x}) + \nabla f(\mathbf{x} + \theta \mathbf{v}) \dfrac{d(\mathbf{x} + \theta \mathbf{v})}{d\theta}$
$= f(\mathbf{x}) + \nabla f(\mathbf{x} + \theta \mathbf{v})\mathbf{v}.$
■

Now suppose $f \in C^2$. Then,

$$f(\mathbf{x} + \mathbf{v}) = f(\mathbf{x}) + \nabla f(\mathbf{x} + \theta \mathbf{v})\mathbf{v} + \frac{1}{2}\mathbf{v}^\top \nabla^2 f(\mathbf{x} + \theta \mathbf{v})\mathbf{v}.$$

1.2.4 Taylor's Theorem

Taylor's general theorem for multivariate functions is beyond the scope of this text, however, we will consider two special cases:

First, suppose $f \in C^1$ is a real multivariate function. Then,

$$f(\mathbf{x}+\mathbf{v}) = f(\mathbf{x}) + \nabla f(\mathbf{x}+\theta\mathbf{v})\mathbf{v} + o(|\mathbf{v}|).$$

Proof:
We have

$$\lim_{|\mathbf{v}|\to 0} \frac{o(|\mathbf{v}|)}{|\mathbf{v}|} = \lim_{\mathbf{v}\to 0} \frac{f(\mathbf{x}+\mathbf{v}) - f(\mathbf{x}) - \nabla f(\mathbf{x})\mathbf{v}}{|\mathbf{v}|}$$

$$= \lim_{\mathbf{v}\to 0} \frac{f(\mathbf{x}+\theta\mathbf{v})\mathbf{v} - \nabla f(\mathbf{x})\mathbf{v}}{|\mathbf{v}|}$$

$$= \lim_{\mathbf{v}\to 0} \frac{\left(\nabla f(\mathbf{x}+\theta\mathbf{v}) - \nabla f(\mathbf{x})\right)\mathbf{v}}{|\mathbf{v}|}$$

$$= \lim_{v\to 0} \left[\left(\nabla f(\mathbf{x}+\theta\mathbf{v}) - \nabla f(\mathbf{x})\right)\frac{\mathbf{v}}{|\mathbf{v}|}\right]$$

$$= 0.$$

■

Now suppose $f \in C^2$. Then,

$$f(\mathbf{x}+\mathbf{v}) = f(\mathbf{x}) + \nabla f(\mathbf{x}+\theta\mathbf{v})\mathbf{v} + \frac{1}{2}\mathbf{v}^\top \nabla^2 f(\mathbf{x}+\theta\mathbf{v})\mathbf{v} + o(|\mathbf{v}|^2).$$

Proof:
We have

$$\lim_{|\mathbf{v}|^2\to 0} \frac{o(|\mathbf{v}|^2)}{|\mathbf{v}|^2} = \lim_{\mathbf{v}\to 0} \frac{f(\mathbf{x}+\mathbf{v}) - f(\mathbf{x}) - \nabla f(\mathbf{x})\mathbf{v} - (1/2)\mathbf{v}^\top \nabla^2 f(\mathbf{x})\mathbf{v}}{|\mathbf{v}|^2}$$

$$= \lim_{\mathbf{v}\to 0} \frac{(1/2)\mathbf{v}^\top \nabla^2 f(\mathbf{x}+\theta\mathbf{v})\mathbf{v} - (1/2)\mathbf{v}^\top \nabla^2 f(\mathbf{x})\mathbf{v}}{|\mathbf{v}|^2}$$

$$= \frac{1}{2}\lim_{\mathbf{v}\to 0} \frac{\mathbf{v}^\top \left(\nabla^2 f(\mathbf{x}+\theta\mathbf{v}) - \nabla^2 f(\mathbf{x})\right)\mathbf{v}}{|\mathbf{v}|^2}$$

$$= \frac{1}{2}\lim_{v\to 0} \left[\frac{\mathbf{v}^\top}{|\mathbf{v}|}\left(\nabla f(\mathbf{x}+\theta\mathbf{v}) - \nabla f(\mathbf{x})\right)\frac{\mathbf{v}}{|\mathbf{v}|}\right]$$

$$= 0.$$

■

1.2.5 Extreme Value Theorem

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is continuous and $K \subseteq \mathbb{R}^n$ is closed and bounded.

Then, $\exists \min_K f$, i.e., a minimum value of f is attained.

2 Convexity

In this section, we will study so-called convex functions.

These functions are of particular interest to us since they have useful properties that simplify optimization.

2.1 Convex Sets

We begin with the notion of a convex set.

2.1.1 The Definition of a Convex Set

A set Ω, is **convex** iff for any $x_1, x_2 \in \Omega$,

$$sx_1 + (1-s)x_2 \in \Omega \forall s \in [0,1].$$

Geometrically, $\{sx_1 + (1-s)x_2, s \in [0,1]\}$ is the set of all points on the line segment between x_1 and x_2, and Ω is convex iff every such line segment is entirely within Ω, i.e., iff

$$\{sx_1 + (1-s)x_2, s \in [0,1]\} \subseteq \Omega.$$

Note that the intersection of two convex sets is also convex.

2.2 Convex Functions

2.2.1 The Definition of a Convex Functions

A function, $f : \Omega \to \mathbb{R}$, where Ω is a convex set is **convex** iff for any $x_1, x_2 \in \Omega$,

$$f(sx_1 + (1-s)x_2) \leq sf(x_1) + (1-s)f(x_2), \forall s \in [0,1].$$

Geometrically, $\{sf(x_1) + (1-s)f(x_2), s \in [0,1]\}$ is the set of all points on the secant line between $f(x_1)$ and $f(x_2)$, and f is convex iff every such secant line is above f.

Alternatively, f is convex iff its epigraph, $\operatorname{epi} f$ is a convex set.

We say f is **strictly convex** iff f is convex and moreover, for any $x_1 \neq x_2 \in \Omega$,

$$f(sx_1 + (1-s)x_2) < sf(x_1) + (1-s)f(x_2), \forall s \in (0,1).$$

Finally, we say f is **concave** (resp. strictly concave) if $-f$ is convex (resp. strictly convex).

2.2.2 Characterizations of Convexity

Let $f \in C^1$ on a convex set, $\Omega \subseteq \mathbb{R}^n$. Then, f is convex if and only if for any $\mathbf{x}, \mathbf{y} \in \Omega$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}).$$

Geometrically, $\{f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})\}$ is the hyperplane tangent to f at $\mathbf{x}$ and f is convex iff all such tangent planes are below it.

Proof:
$\Rightarrow$: Suppose f is convex. Then, by definition, for any $\mathbf{x}, \mathbf{y} \in \Omega$,

$$f(s\mathbf{y} + (1-s)\mathbf{x}) \leq sf(\mathbf{y}) + (1-s)f(\mathbf{x}), \forall s \in [0,1]$$

$$\begin{aligned}
&\Rightarrow f(\mathbf{x} + s(\mathbf{y} - \mathbf{x})) - f(\mathbf{x}) \leq s(f(\mathbf{y}) - f(\mathbf{x})), \forall s \in [0,1] \\
&\Rightarrow \frac{f(\mathbf{x} + s(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{s} \leq (f(\mathbf{y}) - f(\mathbf{x})), \forall s \in (0,1] \\
&\Rightarrow \lim_{s \to 0} \frac{f(\mathbf{x} + s(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{s} \leq \lim_{s \to 0}[(f(\mathbf{y}) - f(\mathbf{x}))], \forall s \in (0,1] \\
&\equiv \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x}) \\
&\Rightarrow f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}).
\end{aligned}$$

$\Leftarrow$: Now suppose $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$ for any $\mathbf{x}, \mathbf{y} \in \Omega$.

Then, consider $\mathbf{z} = s\mathbf{x} + (1-s)\mathbf{y}$ for any $s \in [0,1]$. Since Ω is convex, $\mathbf{z} \in \Omega$, and

$$\begin{aligned}
f(\mathbf{x}) &\geq f(\mathbf{z}) + \nabla f(\mathbf{z})(\mathbf{x} - \mathbf{z}) \\
f(\mathbf{y}) &\geq f(\mathbf{z}) + \nabla f(\mathbf{z})(\mathbf{y} - \mathbf{z})
\end{aligned}.$$

Replacing $\mathbf{z}$ in the above inequalities, we have

$$\begin{aligned}
f(\mathbf{x}) &\geq f(s\mathbf{x} + (1-s)\mathbf{y}) + \nabla f(s\mathbf{x} + (1-s)\mathbf{y})(\mathbf{x} - (s\mathbf{x} + (1-s)\mathbf{y})) \\
f(\mathbf{y}) &\geq f(s\mathbf{x} + (1-s)\mathbf{y}) + \nabla f(s\mathbf{x} + (1-s)\mathbf{y})(\mathbf{y} - (s\mathbf{x} + (1-s)\mathbf{y}))
\end{aligned}.$$

$$\equiv \begin{aligned}
sf(\mathbf{x}) &\geq sf(s\mathbf{x} + (1-s)\mathbf{y}) + s(1-s)\nabla f(s\mathbf{x} + \mathbf{y})(1-s)(\mathbf{x} - \mathbf{y}) \\
(1-s)f(\mathbf{y}) &\geq (1-s)f(s\mathbf{x} + (1-s)\mathbf{y}) + s(1-s)\nabla f(s\mathbf{x} + (1-s)\mathbf{y})(\mathbf{y} - \mathbf{x})
\end{aligned}$$

Adding the above inequalities, we have

$$sf(\mathbf{x}) + (1-s)f(\mathbf{y}) \geq f(s\mathbf{x} + s(1-s)\mathbf{y}), \forall s \in [0,1],$$

which is the definition of convexity. ■

Let $f \in C^2$ on a convex set, $\Omega \subseteq \mathbb{R}^n$. Then, f is convex iff $\nabla^2 f(\mathbf{x}) \succeq 0$, i.e., $\nabla^2 f(\mathbf{x})$ is positive-semi-definite[8].

Proof:
$\Leftarrow$: For any $\mathbf{x}, \mathbf{y} \in \Omega$,

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{y} - \mathbf{x}) + \frac{1}{2}(\mathbf{y} - \mathbf{x})^\top \nabla^2 f(\mathbf{x} + s(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x}).$$

Now if $\nabla^2 f(\mathbf{x}) \succeq 0$, then,

$$\frac{1}{2}(\mathbf{y} - \mathbf{x})^\top \nabla^2 f(\mathbf{x} + s(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x}) \geq 0,$$

and so

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{y} - \mathbf{x}),$$

which as we have already shown is equivalent to f being convex.

$\Rightarrow$: Suppose, towards a contradiction, that f is convex but there exists some $\mathbf{x} \in \Omega$ for which $\nabla^2 f(\mathbf{x}) \not\succeq 0$.

By definition then, there must exist at least one $\mathbf{v} \neq 0 \in \Omega$, such that $\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} < 0$.

Notice that without loss of generality, we may assume $\mathbf{v}$ is arbitrarily small, since $\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} < 0 \Leftrightarrow (k\mathbf{v})^\top \nabla^2 f(\mathbf{x})(k\mathbf{v}) < 0$ for any $k > 0$.

As it turns out, we may also assume, without loss of generality, that $\mathbf{x}$ is an interior point of Ω, i.e., $\mathbf{x} \in \partial\Omega$.

This is because $\nabla^2 f$ is continuous by the assumption that $f \in C^2$, and so there must exist some neighbourhood around $\mathbf{x}$ for which $\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} < 0$.

Thus, we may always chose $\mathbf{v}$ so that $\forall s \in [0, 1]$,

$$\mathbf{v}^\top \nabla^2 f(\mathbf{x} + s\mathbf{v}) \mathbf{v} < 0.$$

Further, if we let $\mathbf{v} = \mathbf{y} - \mathbf{x}$, then

$$(\mathbf{y} - \mathbf{x})^\top \nabla^2 f(\mathbf{x} + s(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x}) < 0,$$

which implies that

$$f(\mathbf{y}) \leq f(x) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}).$$

$\Rightarrow\Leftarrow$ ■

[8] Recall that a matrix, $\mathbf{M}$ is positive semi-definite (resp. positive definite) if for any appropriately sized $\mathbf{v}$, $\mathbf{v}^\top \mathbf{M} \mathbf{v} \geq 0$ (resp. $\mathbf{v}^\top \mathbf{M} \mathbf{v} > 0$).

2.2.3 Properties of Convex Functions

Let Ω be a convex set. We now state a few basic properties of convex functions:

1. If $f_1, f_2 : \Omega \to \mathbb{R}$ are convex, then f defined such that $f(\mathbf{x}) = f_1(\mathbf{x}) + f_2(\mathbf{x})$ is also convex.

   Proof:
   By the definition of f,

   $$f(s\mathbf{x}_1 + (1-s)\mathbf{x}_2) = f_1(s\mathbf{x}_1 + (1-s)\mathbf{x}_2) + f_2(s\mathbf{x}_1 + (1-s)\mathbf{x}_2).$$

   Now since f_1 and f_2 are convex, by definition,

   $$f_1(s\mathbf{x}_1 + (1-s)\mathbf{x}_2) \leq sf_1(\mathbf{x}_1) + (1-s)f_1(\mathbf{x}_2),$$

   and

   $$f_2(s\mathbf{x}_1 + (1-s)\mathbf{x}_2) \leq sf_2(\mathbf{x}_1) + (1-s)f_2(\mathbf{x}_2).$$

   Combining these results we conclude that

   $$f(s\mathbf{x}_1 + (1-s)\mathbf{x}_2) \leq s(f_1(\mathbf{x}_1) + f_2(\mathbf{x}_1)) + (1-s)(f_1(\mathbf{x}_2) + f_2(\mathbf{x}_2)))$$

   $$= sf(\mathbf{x}_1) + (1-s)f(\mathbf{x}_2),$$

   which implies that f is convex. ■

2. If f is convex, and $a \in \mathbb{R}_+$, then g defined such that $g(\mathbf{x}) = af(\mathbf{x})$ is convex.

   Proof:
   By the definition of g,

   $$g(s\mathbf{x}_1 + (1-s)\mathbf{x}_2) = af(s\mathbf{x}_1 + (1-s)\mathbf{x}_2).$$

   Now since f is convex, by definition,

   $$f(s\mathbf{x}_1 + (1-s)\mathbf{x}_2) \leq sf(\mathbf{x}_1) + (1-s)f(\mathbf{x}_2).$$

   Combining these results we conclude that

   $$g(s\mathbf{x}_1 + (1-s)\mathbf{x}_2) \leq saf(\mathbf{x}_1) + (1-s)af(\mathbf{x}_2)$$

   $$= sg(\mathbf{x}_1) + (1-s)g(\mathbf{x}_2),$$

   which implies that g is convex. ■

3. If f is convex, the set $\Gamma_{f,k} = \{\mathbf{x} \in \Omega | f(\mathbf{x}) \leq k\}$ is convex.

   Proof:
   Fix $\mathbf{x}_1, \mathbf{x}_2 \in \Gamma^-_{f,k}$. Since f is convex, by definition,

   $$f(s\mathbf{x}_1 + (1-s)\mathbf{x}_2) \leq sf(\mathbf{x}_1) + (1-s)f(\mathbf{x}_2), s \in [0,1]$$

   $\leq sk + (1-s)k$
   $\leq k$

   Thus, $s\mathbf{x}_1 + (1-s)\mathbf{x}_2 \in \Gamma^-_{f,k}$, for every $s \in [0,1]$.

Part II
Finite Dimensional Optimization

Given a function, $f : \Omega \to \mathbb{R}$, optimization deals with finding an extrema of f, say $f(\mathbf{x}^*)$ typically so that $\mathbf{x}^*$ satisfies a particular set of constraints.

In particular, we will be studying optimization problems of the form

$$\begin{aligned} \underset{\Omega}{\text{minimize}} \quad & f_0(\mathbf{x}) \\ \text{subject to} \quad & f_i(\mathbf{x}) \leq b_i, \; i = 1, \ldots, m, \end{aligned}$$

where, $f_0 : \Omega \to \mathbb{R}$ is called the **objective function**, and $f_i : \Omega \to \mathbb{R}$ for $i = 1, \ldots, m$ are the **constraints**[9].

2.2.4 The Relationship between Minimization and Maximization

Our first goal is to show that we may always frame a minimization problem in terms of a maximization problem and visa-versa.

In particular, for any $f : \Omega \to \mathbb{R}$. Then, for any $K \subseteq \Omega$, $\max_K f = \min_K(-f)$.

Proof:
Suppose $\mathbf{x}^*$ is a maximizer for f on $K \subseteq \Omega$, that is, for every $\mathbf{x} \in K$,

$$f(\mathbf{x}) \leq f(\mathbf{x}^*).$$

Then, clearly $-f(\mathbf{x}^*) \leq -f(\mathbf{x})$, meaning $\mathbf{x}^*$ is a minimizer of $-f$ on K.

We can analogously show the converse also holds. ■

For this reason, we will explicitly focus on minimization problems.

[9] We may always express any equality constraint, i.e., that of the form $f_i(\mathbf{x}) = b_i$, by using a pair of inequality constraints, namely, $f_i(\mathbf{x}) \leq b_i$ and $-f_i(\mathbf{x}) \leq -b_i$.

3 Unconstrained Optimization

In this section, we study optimization problems when no constraints are involved. First, we define the concept of a local minimum. Then, we explain how local minima can be used to find global minima, i.e., a solution to the optimization problem.

3.0.1 The Definition of Local Minimum

We say $\mathbf{x}^* \in \Omega$, is a **local minimizer** of a function, $f : \Omega \to \mathbb{R}$, iff there exists some neighbourhood, N, around $\mathbf{x}^*$, such that for every $\mathbf{x} \in N$,

$$f(\mathbf{x}^*) \leq f(\mathbf{x}).$$

We refer to $f(\mathbf{x}^*)$ as a **local minimum**.

If $\mathbf{x}*$ is a local minimizer of f and $f(\mathbf{x}) < f(\mathbf{x}^*)$ for every $\mathbf{x} \neq \mathbf{x}^* \in N$, then $f(\mathbf{x}^*)$ is a **strict local minimim** and $\mathbf{x}^*$ is a **strict local minimizer**.

3.1 Conditions for a Local Minimum

Now we study the necessary and sufficient conditions for local minima.

3.1.1 Necessary Conditions for a Local Minimum

Consider some function, $f : \Omega \to \mathbb{R}$.

Suppose $f \in C^1$. Then, for any local minimizer, $\mathbf{x}^*$, it holds that

$$\nabla f(\mathbf{x}^*) \cdot \mathbf{v} \geq 0,$$

for any feasible $\mathbf{v}$.

Proof:
For any feasible, $\mathbf{v}$, let $g(s) := f(\mathbf{x}^* + s\mathbf{v})$.

Since we assume $\mathbf{x}^*$ is a local minimizer of f on Ω, by definition, $\exists \bar{s} > 0$, so that $f(\mathbf{x}^*) \leq f(\mathbf{x}^* + s\mathbf{v}), \forall s \in [0, \bar{s}]$.

Equivalently, we have $g(0) \leq g(s), \forall s \in [0, \bar{s}]$, which means $s = 0$ is a local minimizer of g, which means $g'(0) \geq 0$. But,

$$g'(0) = \frac{d}{ds}\bigg|_{s=0} g(s) = \frac{d}{ds}\bigg|_{s=0} f(\mathbf{x}^* + s\mathbf{v}) = \nabla f(\mathbf{x}^*) \cdot \mathbf{v} \Rightarrow \nabla f(\mathbf{x}^*) \cdot v \geq 0,$$

as required. ■.

As a corollary, if $\mathbf{x}^*$ is an interior point of Ω, i.e., $\mathbf{x}^* \in \Omega \setminus \partial\Omega$, then every direction is feasible and so it must hold that

$$\nabla f(\mathbf{x}^*) = 0.$$

Now suppose $f \in C^2$. Then, for any local minimizer, $\mathbf{x}^*$, if $\mathbf{v}$ is feasible, then

$$\nabla f(\mathbf{x}^*) \cdot \mathbf{v} = 0 \Rightarrow \mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} \geq 0.$$

Proof:
Fix $\mathbf{v}$. Using the Taylor approximation for f, we have

$$f(\mathbf{x}^* + s\mathbf{v}) - f(\mathbf{x}^*) = s\nabla f(\mathbf{x}^*) \cdot \mathbf{v} + \frac{1}{2}s^2\mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} + o(s^2).$$

Since we assume $\mathbf{x}^*$ is a local minimizer of f on $\mathbb{R}^n$, by definition, $\exists \bar{s} > 0$, so that $f(\mathbf{x}^*) \leq f(\mathbf{x}^* + s\mathbf{v}), \forall s \in [0, \bar{s}]$.

Therefore, for any $s \in [0, \bar{s}]$,

$$s\nabla f(\mathbf{x}^*) \cdot \mathbf{v} + \frac{1}{2}s^2\mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} + o(s^2) \geq 0.$$

Now by assumption, $\nabla f(\mathbf{x}^*) \cdot \mathbf{v} = 0$, and so

$$\frac{1}{2}s^2\mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} + o(s^2) \geq 0$$

$$\Rightarrow \frac{1}{2}\mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} + \frac{o(s^2)}{s^2} \geq 0, s \neq 0$$

Notice that the above inequality must hold $\forall s \in (0, \bar{s}]$, and in particular, it should hold in the limit as $s \to 0$. Thus,

$$\lim_{s \to 0} \frac{1}{2}\mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} + \frac{o(s^2)}{s^2} \geq 0, s \neq 0$$

$$\Rightarrow \frac{1}{2}\mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} \geq 0.$$

■

As a corollary, if $\mathbf{x}^*$ is an interior point of Ω, i.e., $\mathbf{x}^* \in \Omega \setminus \partial\Omega$, then every direction is feasible and so it must hold that

$$\nabla^2 f(\mathbf{x}^*) \succeq 0.$$

3.1.2 Sufficient Conditions for a Local Minimum

Consider some function, $f : \Omega \to \mathbb{R}$, and some $\mathbf{x}_0$ in the interior of Ω (we will consider the case where $\mathbf{x}_0$ is a boundary point after).

If $\nabla f(\mathbf{x}_0) = 0$ and $\nabla^2 f(\mathbf{x}_0) \succ 0$, then $\mathbf{x}_0$ is a (strict) local minimizer.

Proof:
Using the Taylor approximation for f, we have

$$f(\mathbf{x}^* + \mathbf{v}) - f(\mathbf{x}^*) = \nabla f(\mathbf{x}^*) \cdot \mathbf{v} + \frac{1}{2}\mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} + o(|\mathbf{v}|^2).$$

Since we assume $\mathbf{x}^*$ is a local minimizer of f on $\mathbb{R}^n$, a necessary condition is that $\nabla f(\mathbf{x}^*) \cdot \mathbf{v} = 0$.

$$\therefore f(\mathbf{x}^* + \mathbf{v}) - f(\mathbf{x}^*) = \frac{1}{2}\mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} + o(|\mathbf{v}|^2)$$

$$\geq |\mathbf{v}|^2 \left(\frac{a}{2} + \frac{o(|\mathbf{v}|^2)}{|\mathbf{v}|^2}\right) \because \exists a \text{ s.t. } \mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} \geq a|\mathbf{v}|^2, \forall \mathbf{v}$$

To see the latter inequality, we first observe that since $\nabla^2 f(\mathbf{x}^*)$ is symmetric, it is orthogonally diagonalizable, that is,

$$\nabla^2 f(\mathbf{x}^*) = Q^\top D Q,$$

where D is a diagonal matrix, and Q is an orthogonal matrix.

$$\begin{aligned}
&\Rightarrow \mathbf{v}^\top \nabla^2 f(\mathbf{x}^*)\mathbf{v} = \mathbf{v}^\top Q^\top D Q \mathbf{v} \\
&= (Q\mathbf{v})^\top D (QV) \\
&= \mathbf{w}^\top D \mathbf{w} \\
&= \begin{bmatrix} w_1 & \dots & w_n \end{bmatrix} \begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} \\
&= \sum_{i=1}^{n} \lambda_i w_i^2 \\
&\geq a \sum_{i=1}^{n} w_i^2, a = \min(\{\lambda_1, \dots, \lambda_n\}) \\
&= a|\mathbf{w}|^2 \\
&= a|\mathbf{v}|^2.
\end{aligned}$$

Note that the last equality follows from the fact that orthogonal matrices preserve norms.

Now, since we assumed $\nabla^2 f(\mathbf{x}^*) \succ 0$, it must be that $a > 0$. This is because every eigenvalue of a positive-definite matrix must be positive.

Thus, for small enough $\mathbf{v}$,

$$\frac{a}{2} + \frac{o(|\mathbf{v}|^2)}{|\mathbf{v}|^2} > 0,$$

meaning

$$f(\mathbf{x}^* + \mathbf{v}) - f(\mathbf{x}^*) > 0 \Rightarrow f(\mathbf{x}^*) < f(\mathbf{x}^* + \mathbf{v}),$$

again, for sufficiently small $\mathbf{v}$.

Thus, $\mathbf{x}^*$ is a strict local minimizer. ■

3.2 Conditions for a Global Minimum

Now we can study how local minima relate to global minima.

3.2.1 The Definition of a Global Minimum

We say $\mathbf{x}^* \in \Omega$ is a **global minimizer** of a function, $f : \Omega \to \mathbb{R}$, iff for every $\mathbf{x} \in \Omega$,

$$f(\mathbf{x}^*) \leq f(\mathbf{x}).$$

We can easily show that any global minimizer must be a local minimizer.

Proof:
Assume towards a contradiction that $\mathbf{x}^*$ is a global minimizer of f but not a local minimizer.

Then, by definition, for every neighbourhood N, around $\mathbf{x}^*$, there exists at least one $\mathbf{x}' \neq \mathbf{x}^*$ such that

$$f(\mathbf{x}^*) > f(\mathbf{x}') \Leftrightarrow f(\mathbf{x}') < f(\mathbf{x}^*).$$

But this means that $\mathbf{x}^*$ cannot be a global minimum according to the definition. $\Rightarrow\Leftarrow$ ■

Thus, we may always find the global minima of a function by finding the minima of the local minima.

3.2.2 Global Minima of Convex Functions

Every local minimum of a convex function, $f : \Omega \to \mathbb{R}$ is a global minimum.

Proof:
Assume towards a contradiction that $\mathbf{x}^*$ is a local minimizer of f, but not a global minimizer.

Then, there must exist some $\mathbf{y} \in \Omega$, such that $f(\mathbf{y}) < f(\mathbf{x}^*)$.

By convexity, for any $s \in [0, 1]$,

$$f(s\mathbf{x}^* + (1-s)\mathbf{y}) \leq sf(\mathbf{x}^*) + (1-s)f(\mathbf{y}),$$

$< sf(\mathbf{x}^*) + (1-s)f(\mathbf{x}^*)$
$= f(\mathbf{x}^*)$
which means that $\mathbf{x}^*$ cannot be a local minimizer. ■

Now, suppose $\min_\Omega f = m$ and let $\Gamma_{f,m}$ be the set of global minimizers for f, that is,

$$\Gamma_{f,k} := \{\mathbf{x} \in \Omega | f(\mathbf{x}) = \min_\Omega f\} \neq \emptyset.$$

Then, $\Gamma_{f,m}$ is a convex set. Thus, if $\mathbf{x}_1, \mathbf{x}_2$ are global minimizers, then $s\mathbf{x}_1 + (1-s)\mathbf{x}_2, s \in [0,1]$ are all also global minimizers.

Proof:
Note $\Gamma^-_{f,k} = \{\mathbf{x} \in \Omega | f(\mathbf{x}) \leq m\} = \{\mathbf{x} \in \Omega | f(\mathbf{x}) = m\} \cup \{\mathbf{x} \in \Omega | f(\mathbf{x}) < m\}$.

But, by the definition of a global minimum, $\nexists \mathbf{x}$ s.t. $f(\mathbf{x}) < \min_\Omega f$, so $\{\mathbf{x} \in \Omega | f(\mathbf{x}) < m\} = \emptyset$.

$$\therefore \Gamma_{f,k} = \{\mathbf{x} \in \Omega | f(\mathbf{x}) \leq m\} = \Gamma^-_{f,k},$$

which we have already shown to be convex.

If Ω is also closed and bounded, we can show that $\max_\Omega f = \max \partial\Omega f$.

Proof:
Since Ω is closed ($\partial\Omega \neq \emptyset$), it follows that $\partial\Omega \subseteq \Omega$, and so

$$\max_\Omega f \geq \max_{\partial\Omega} f.$$

4 Optimization with Equality Constraints